

T:- 98, 104, 118,119



Domain:- Web development

Project title:- Smart Attendance Management System with
Facial Recognition and Performance Analytics

Team members

Preethika.C - PES1UG24CA119

Preethi - PES1UG24CA118

Nesara - PES1UG24CA104

Monisha - PES1UG24CA098

Guide

Prof. Niteesh K R

Problem Statement



- Traditional attendance systems rely on manual or basic digital methods that are prone to errors, proxy attendance, and lack detailed tracking of student participation.
- These systems fail to record late entry, early exit, actual time spent in class, and do not analyze the relationship between attendance and academic performance.
- Hence, there is a need for a secure, automated, and intelligent attendance management system.

Motivation:-

- Traditional attendance systems rely on manual registers or simple digital entry, which are time-consuming and error-prone.
- Proxy attendance is common when students mark attendance for absent classmates.
- Faculty spend significant time recording and managing attendance data instead of focusing on teaching.
- Current systems often do not track actual classroom presence duration, late entry, or early exit.
- Parents and administrators usually lack real-time visibility of student attendance.
- There is also no automatic way to correlate attendance with academic performance.

Therefore, an intelligent and automated attendance solution is required that ensures accuracy, transparency, and efficiency.

Motivation and Needs

Needs:-

- Automated attendance recording using Facial Recognition and QR-based Digital ID Cards.
- Elimination of proxy attendance through Geofencing and location validation.
- Late entry and early exit tracking to measure actual classroom participation time.
- Real-time notifications to parents via SMS when students enter or miss classes.
- Secure performance tracking, where only lecturers can update student marks.
- Attendance vs Performance correlation to identify patterns affecting academic results.
- Map-based attendance visualization for administrators to monitor attendance locations.
- Remote access through web or mobile interfaces for easy monitoring.

This system helps educational institutions improve attendance accuracy, reduce administrative workload, enhance student accountability, and support better academic decision-making.

Problem Definition:-



- Traditional attendance systems are largely manual or semi-automated, making them prone to errors, manipulation, and proxy attendance.
- These systems fail to accurately verify physical presence and do not capture detailed participation metrics such as time spent in class, late arrivals, or early departures.
- Moreover, existing systems lack analytical capabilities to evaluate the impact of attendance on academic performance.
- The absence of location validation and real-time monitoring reduces the reliability and usefulness of attendance data.
- Hence, there is a need for a secure, automated, and analytics-driven attendance management system.

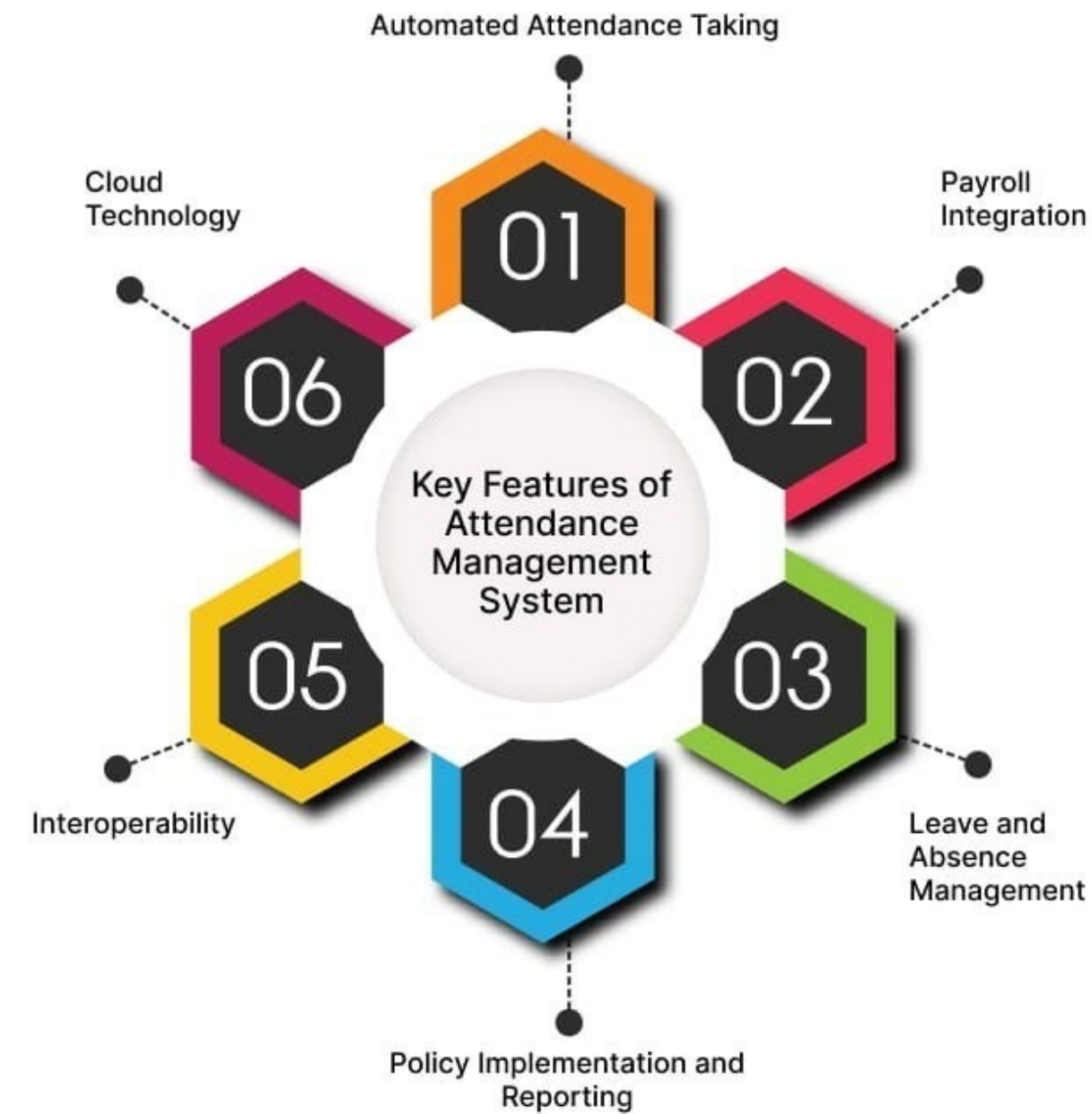
Objectives:-



- To automate attendance recording using facial recognition and digital ID authentication.
- To eliminate proxy attendance through geo-fencing and location validation.
- To track late entry, early exit, and total classroom duration.
- To analyze attendance data and correlate it with academic performance.
- To generate real-time attendance reports and dashboards

Scope of the solution:-

- Applicable to educational institutions and organizations.
- Supports real-time attendance tracking and reporting.
- Provides performance analysis based on attendance data.
- Scalable to support large numbers of users.
- Can be extended with mobile and cloud-based features



Users of the application

- Students / Employees – to mark attendance and view records.
- Faculty / Supervisors – to monitor attendance and performance.
- Administrators – to manage users, reports, and system settings



Domain Study:-



- This project falls under the domains of Web Application Development, Computer Vision, and Data Analytics.
- It utilizes facial recognition technology for identity verification, location-based services for geo-fencing, and analytical techniques to evaluate attendance and performance data.
- The system integrates modern technologies to enhance accuracy, security, and efficiency in attendance management.

- Attendance tracking is important in schools, colleges, and offices.
- Traditional methods use manual registers and roll calls.
- Modern systems use RFID, biometrics, QR codes, GPS, and facial recognition.
- These technologies improve accuracy, automation, and efficiency.
- However, many systems still face issues like proxy attendance and hardware dependency.

Literature overview



Number of papers reviewed

Total papers reviewed: 15

Focus Area:-

- CNN based face recognition attendance system. Real time face detection classroom
- QR code based attendance system
- Geo-fencing attendance system using GPS
- Time tracking student attendance system
- Attendance and academic performance correlation analysis

Year Distribution:-2022(4 papers),
2023(3 papers), 2024(1 paper),
2025(5 paper), 2026(2 paper)

Selection Criteria:-

The papers were selected based on:-

- Relevance to face recognition
- Published in IEEE/ reputed international conferences
- Reported performance metrics(accuracy, security, efficiency, real-time monitoring, scalability)

Research Paper 1



Authors / Published Year / Citation:- Mijanur Rahman, Md. Masudul Islam, Dip Nandi

Published 2025 – Journal of Electrical Systems and Information Technology

Objective:- To develop a Wi-Fi based smart attendance system (SmartPresence) that automatically records student attendance using Wi-Fi signals and mobile devices.

Methodology

- Uses Wi-Fi signals and MAC address detection from student smartphones.
- Attendance captured automatically through network connectivity.
- Data stored in a database and web application for report generation.
- System architecture includes data collection, IoT device integration, and web-based monitoring.

Contribution

- Introduced Wi-Fi based automated attendance tracking.
- Provides real-time attendance monitoring.
- Reduces manual work and administrative effort.

Strengths

- No biometric hardware required.
- Cost-effective using existing Wi-Fi infrastructure.
- Real-time attendance tracking.

Limitations

- Requires students to carry Wi-Fi enabled smartphones.
- Accuracy depends on Wi-Fi signal strength and network reliability.

Relevance to Your Project:- This paper is relevant to our project because it demonstrates how automated technologies can be used to improve attendance management by reducing manual work and generating digital attendance records.

Authors / Published Year / Citation

Subhan Behera, Jitendra Kumar Das, Balaram Sahoo

Published 2025 – Journal of Advance and Future Research

Objective

To design an IoT-based smart attendance system using facial recognition and cloud integration for real-time monitoring.

Methodology

- Camera connected to Raspberry Pi for capturing images.
- Facial recognition algorithms used for identifying users.
- Attendance data stored in cloud platforms like Firebase.
- Admin dashboard allows real-time monitoring and report generation.

Contribution

- Combines IoT devices, facial recognition, and cloud storage.
- Provides real-time data synchronization and remote monitoring.

Strengths

- High accuracy (~95%).
- Remote monitoring through web dashboard.
- Secure cloud storage with encryption.

Limitations

- Requires good lighting conditions for facial recognition.
- Hardware cost (camera + Raspberry Pi).

Relevance to Your Project:- This paper is relevant to our project because it shows how IoT and facial recognition technologies can automate attendance tracking and provide real-time monitoring and accurate attendance records.

Research paper 3



Authors / Published Year / Citation

Rupam Sau, Anwesha Bandyopadhyay, Sourav Sarkar, Rohit Kumar Guha, Maitreyee Ganguly

Published 2022 – IEEE Conference on Computational Intelligence and Networks (CINE)

Objective

To create a smart attendance and performance monitoring system using CCTV cameras and AI-based face recognition.

Methodology

- CCTV cameras capture video at entrances.
- Deep Neural Network (DNN) based face detection used.
- Techniques used:
 - Histogram of Oriented Gradients (HOG)
 - Euclidean distance
 - Face encoding
- System analyzes employee movements and generates reports.

Contribution

- Integrates attendance tracking with employee performance monitoring.
- Uses existing CCTV infrastructure for automated monitoring.

Strengths

- No manual interaction required.
- Detects unauthorized individuals.
- Tracks employee activity and attendance.

Limitations

- Requires multiple CCTV cameras.
- High computational processing.

Relevance to Your Project:- This paper is relevant to our project because it explains how artificial intelligence and face recognition can be used to automatically identify individuals and record attendance without manual intervention.

Research Paper 4



Authors / Published Year / Citation

Published 2022 – Advances in Mathematical & Computational Sciences Journal

Objective

To develop a mobile-based attendance monitoring system using facial recognition and geofencing technology.

Methodology

- Android application with OpenCV facial recognition.
- Uses geofencing technology to detect location.
- Attendance recorded only when:
 - Face is recognized
 - Student is inside the defined location boundary.

Contribution

- Combines location tracking (GPS) and facial recognition.
- Reduces proxy attendance and ensures student presence.

Strengths

- Prevents fake attendance.
- Mobile-based and easy to use.

Limitations

- Requires GPS accuracy.
- Depends on smartphone availability.

Relevance to Your Project: This paper is relevant to our project because it highlights how mobile technologies and facial recognition can help in developing a secure and automated digital attendance system.

Research Paper 5



Authors / Published Year / Citation

Bawar Ali Abdalkarim, Devrim Akgün

Published 2022 – 3rd International Conference on Applied Engineering and Natural Sciences (ICAENS)

Objective

To review and analyze different smart attendance systems developed using modern technologies such as biometric recognition, QR codes, GPS, RFID, and facial recognition.

Methodology

- Conducts a literature review of various automated attendance systems.
- Compares technologies like:
 - Face recognition
 - Fingerprint recognition
 - QR code scanning
 - RFID
 - GPS-based attendance
 - Mobile applications.

Contribution

- Provides a comprehensive overview of different smart attendance technologies.
- Identifies advantages, challenges, and trends in automated attendance systems.

Strengths

- Covers multiple attendance technologies in one study.
- Helps researchers understand existing solutions and improvements.
- Provides comparison of different methods.

Limitations

- It is a review paper, so it does not implement a new system.
- Limited practical evaluation or experimental results.

Relevance to Your Project

This paper is relevant to our project because it provides an overview of different smart attendance technologies and highlights how automated systems can improve accuracy, reduce manual work, and simplify attendance management.

Comparative summary table

Functionality / Feature	Paper 1 Year(2025)	Paper 2 Year (2025)	Paper 3 Year(2022)	Paper 4 Year(2022)	Paper 5 Year(2022)
Feature A (What is usable?)	Automated attendance using Wi-Fi and mobile device detection	IoT-based attendance with facial recognition and cloud storage	AI-based face recognition using CCTV cameras	Mobile attendance with facial recognition and geofencing	Review of different smart attendance technologies such as RFID, QR, face recognition
Algorithm Used	Network detection and device identification through MAC address	OpenCV face detection and FaceNet / Dlib recognition algorithms	Deep Neural Network (DNN) with HOG and face encoding	OpenCV facial recognition with GPS location verification	Multiple algorithms discussed such as LBPH, Haar classifier, QR scanning and biometric algorithms
Method Adopted	Detects student presence through Wi-Fi connectivity	Raspberry Pi captures images and sends recognized data to cloud database	CCTV cameras capture faces and AI identifies individuals automatically	Mobile app captures face and verifies user location	Literature review and comparison of existing smart attendance methods
Evaluation Metrics (Accuracy / Dataset)	Evaluated based on successful detection of connected devices	Achieved around 95% accuracy in face recognition	Evaluated based on face detection accuracy and monitoring capability	Accuracy measured based on correct facial and location verification	Comparison based on technologies and performance discussed in previous studies
Limitations (Identified Gaps)	Depends on Wi-Fi availability and smartphone presence	Requires hardware setup and proper lighting for face recognition	Requires multiple cameras and high computational power	Depends on GPS accuracy and smartphone availability	No practical implementation; only reviews existing technologies

Proposed Solution:-



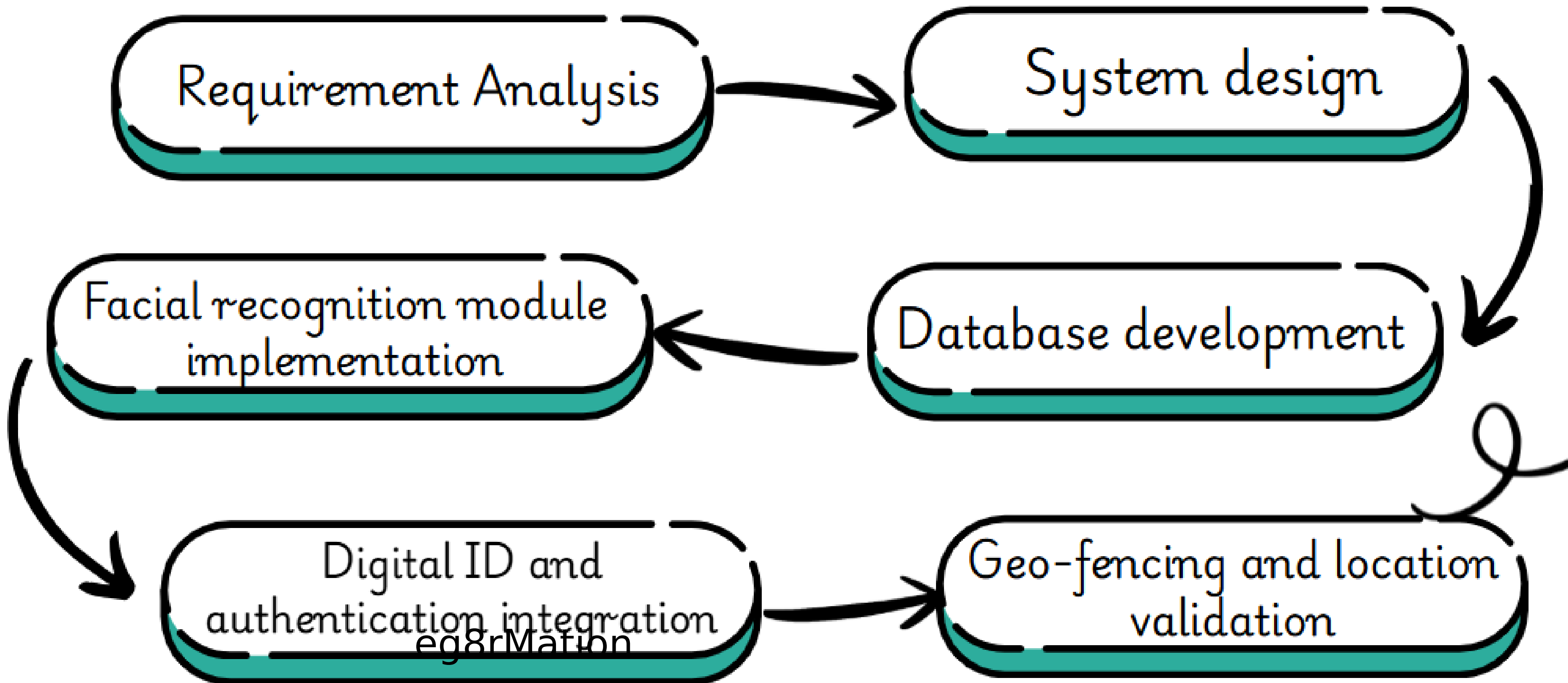
- The proposed system implements an automated attendance mechanism using facial recognition or digital ID verification within a geo-fenced environment.
- Attendance is recorded only when the user is physically present at the designated location.
- The system logs entry and exit times to calculate attendance duration and generates analytical reports that compare attendance patterns with academic performance.

Functionality

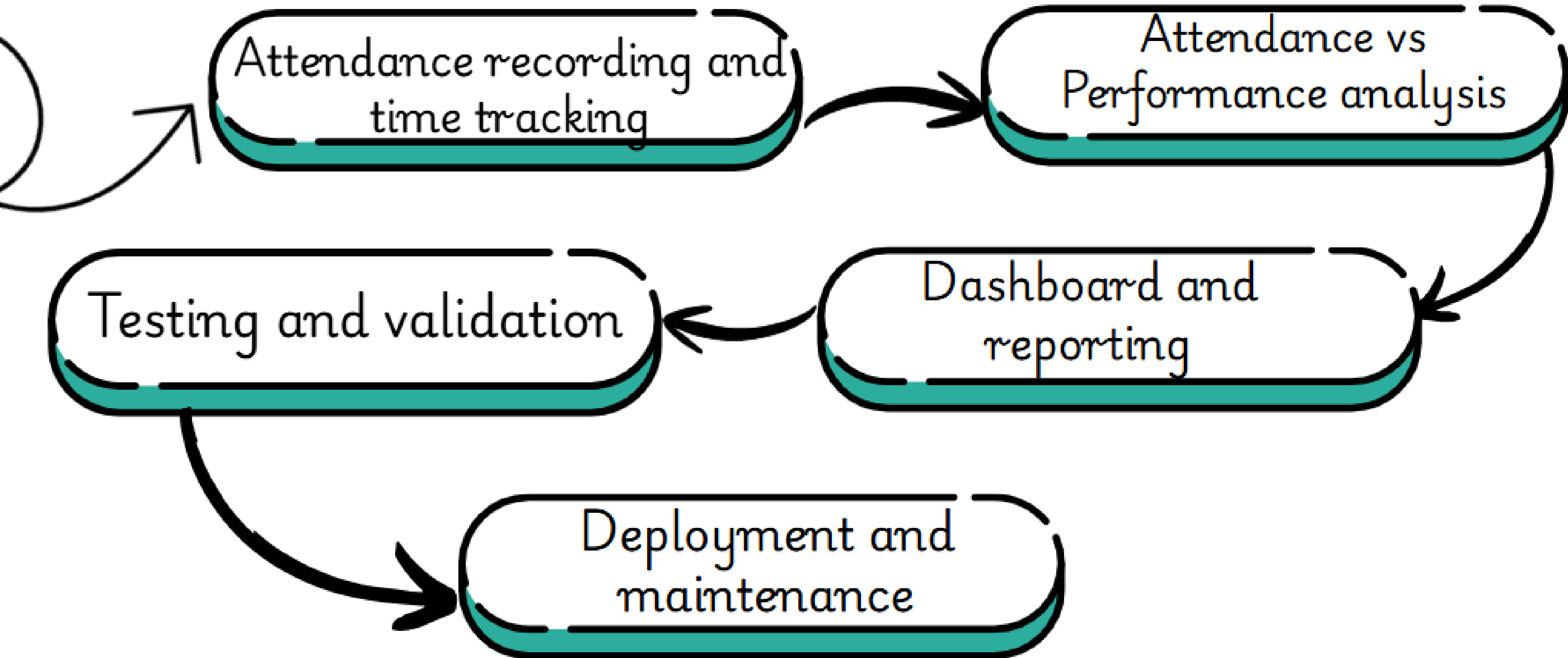


- Facial recognition-based attendance marking.
- Digital ID card authentication
- Geo-fenced location validation to prevent proxy attendance
- Late entry and early exit detection
- Calculation of total attendance duration
- Attendance vs performance correlation analysis
- Real-time reports and dashboards

Steps



Steps (Continue)



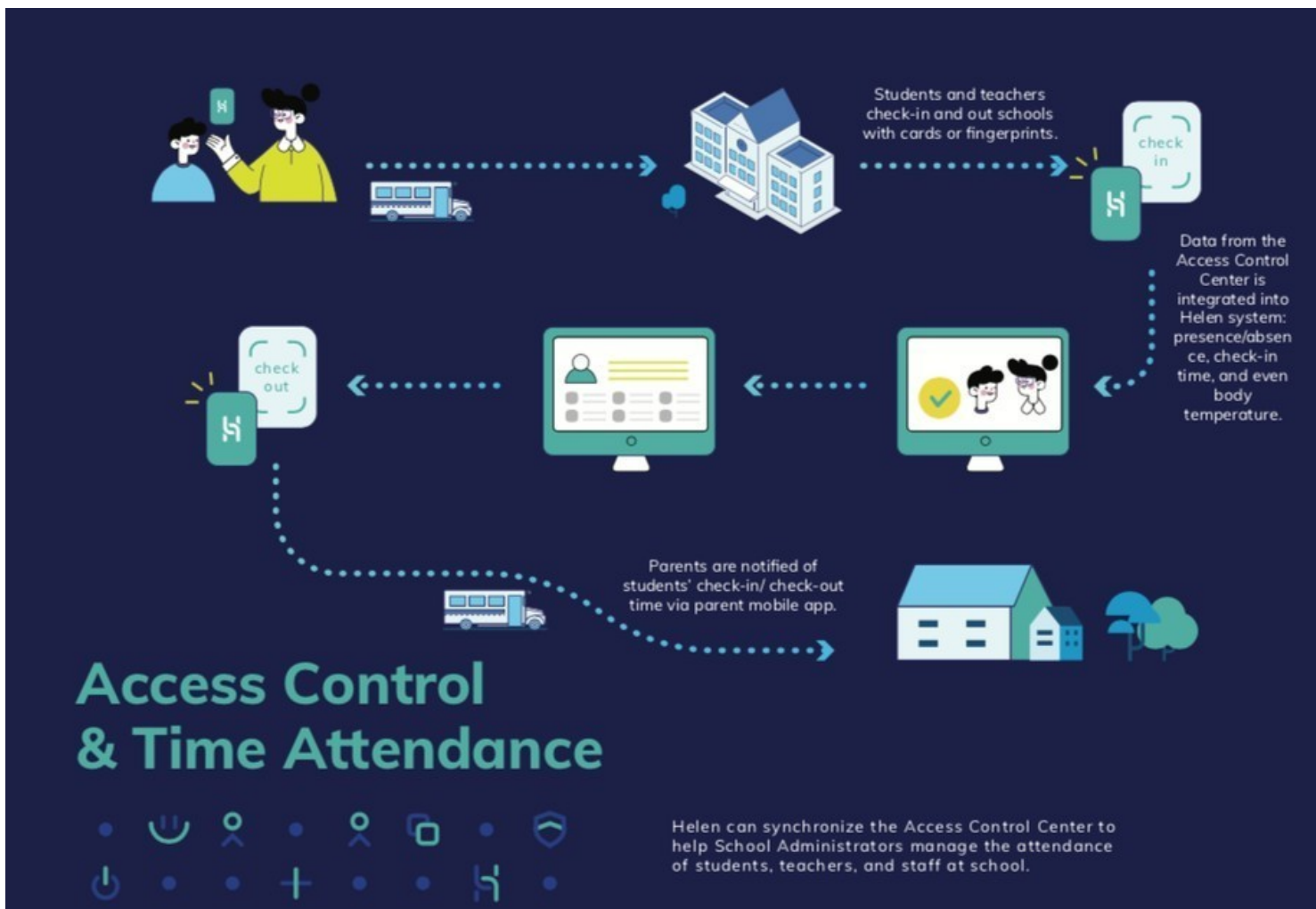
Technologies used



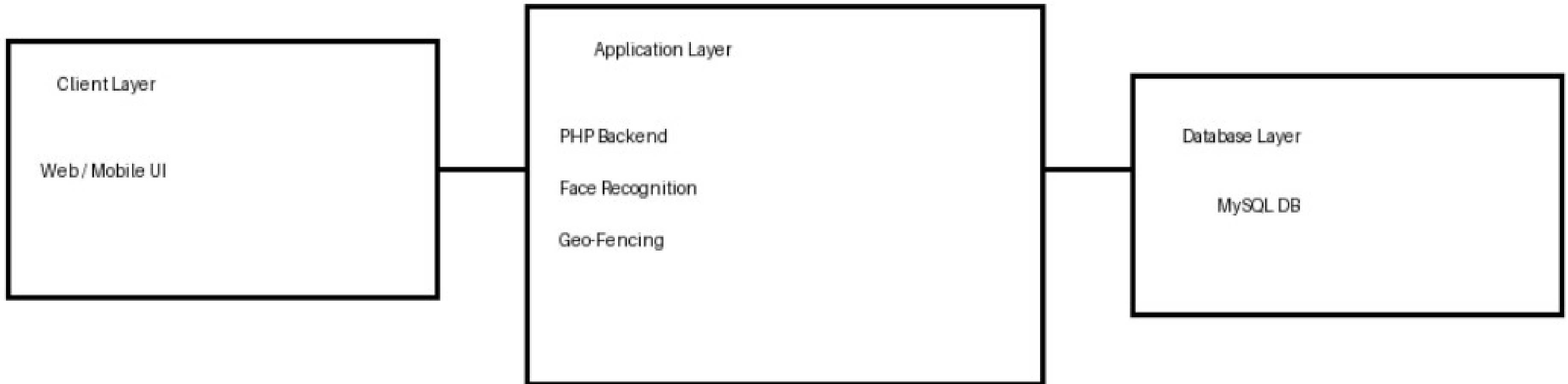
- OpenCV:-Used for face detection and face recognition.
- Firebase:-Used as a cloud database.
- Android studio:-Used to develop mobile applications (Android apps).Can integrate camera, QR scanning, and attendance tracking.
- Visual studio code:-Used for coding the backend or web interface.
- MySQL:-Used as a database to store attendance records.
- Flask:-Used to build the web dashboard.
- AppSheet:-A no-code platform to create mobile or web apps without much programming.
- Adobe Dreamweaver:- Used for front end design

Expected Deliverables

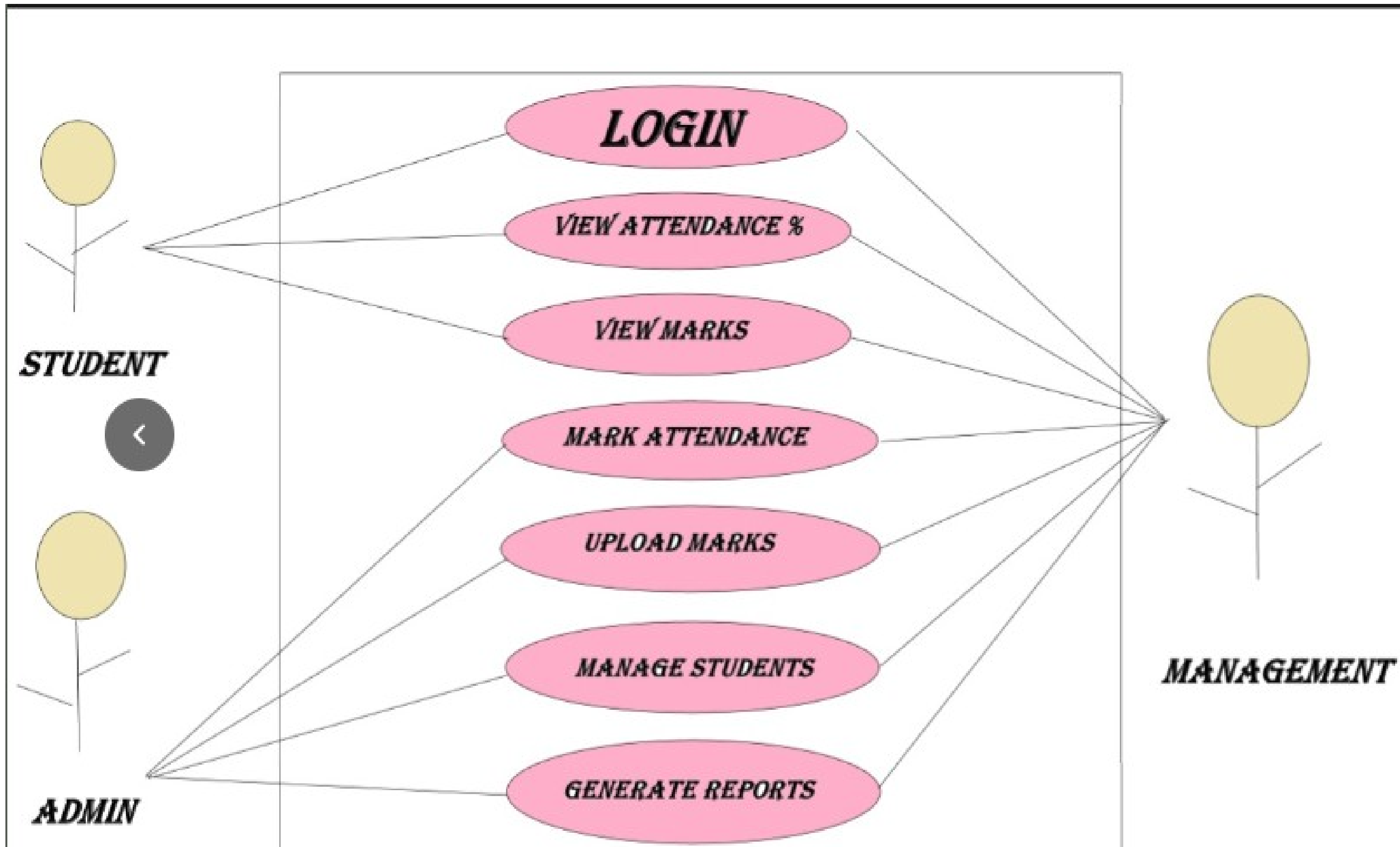
- Accurate and tamper-proof attendance records.
- Complete elimination of proxy attendance.
- Detailed tracking of student classroom participation time.
- Clear insights into the relationship between attendance and academic performance.
- Reduced administrative workload and improved efficiency
- Enhanced data-driven academic monitoring and decision-making.



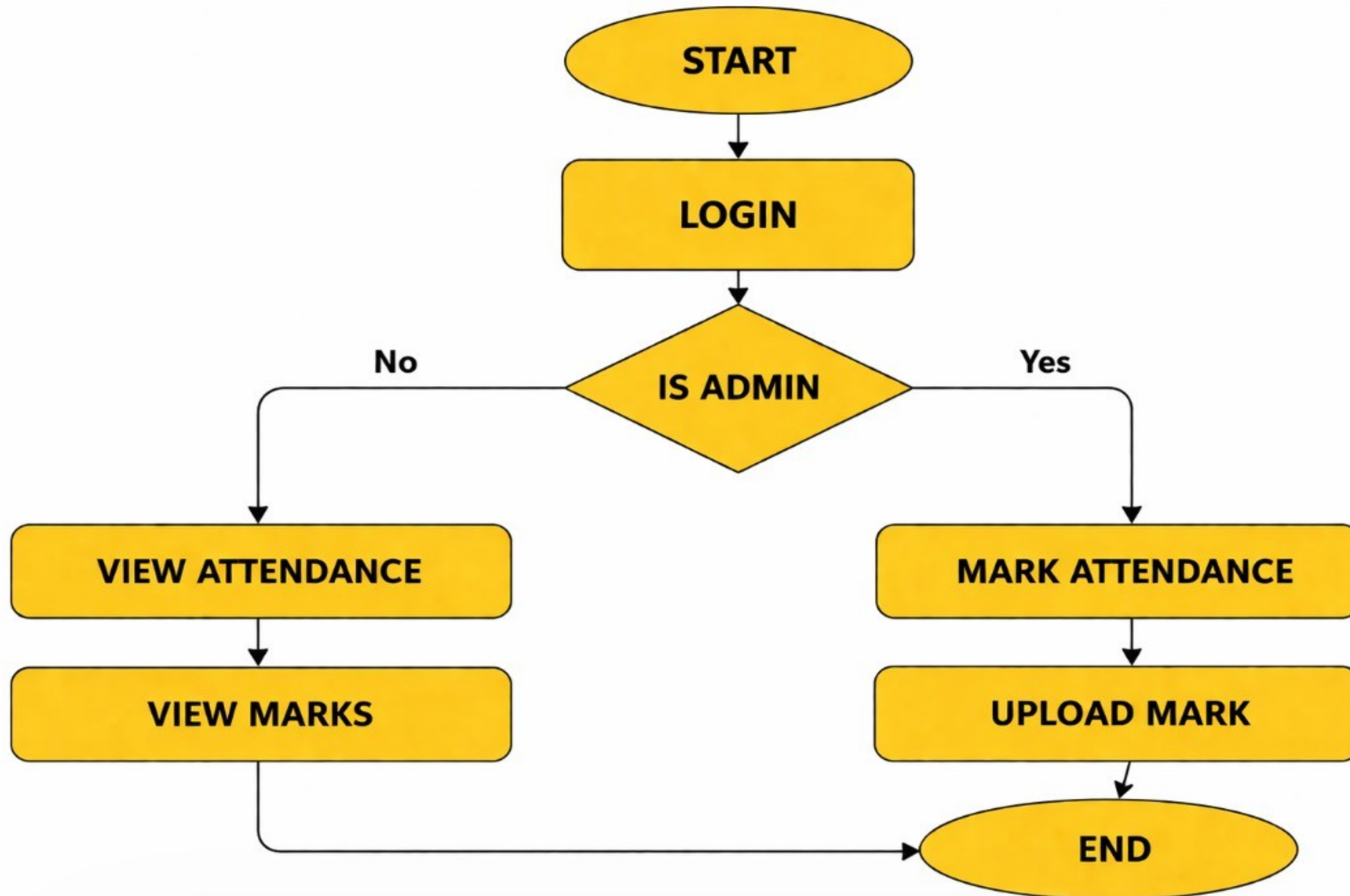
System architecture



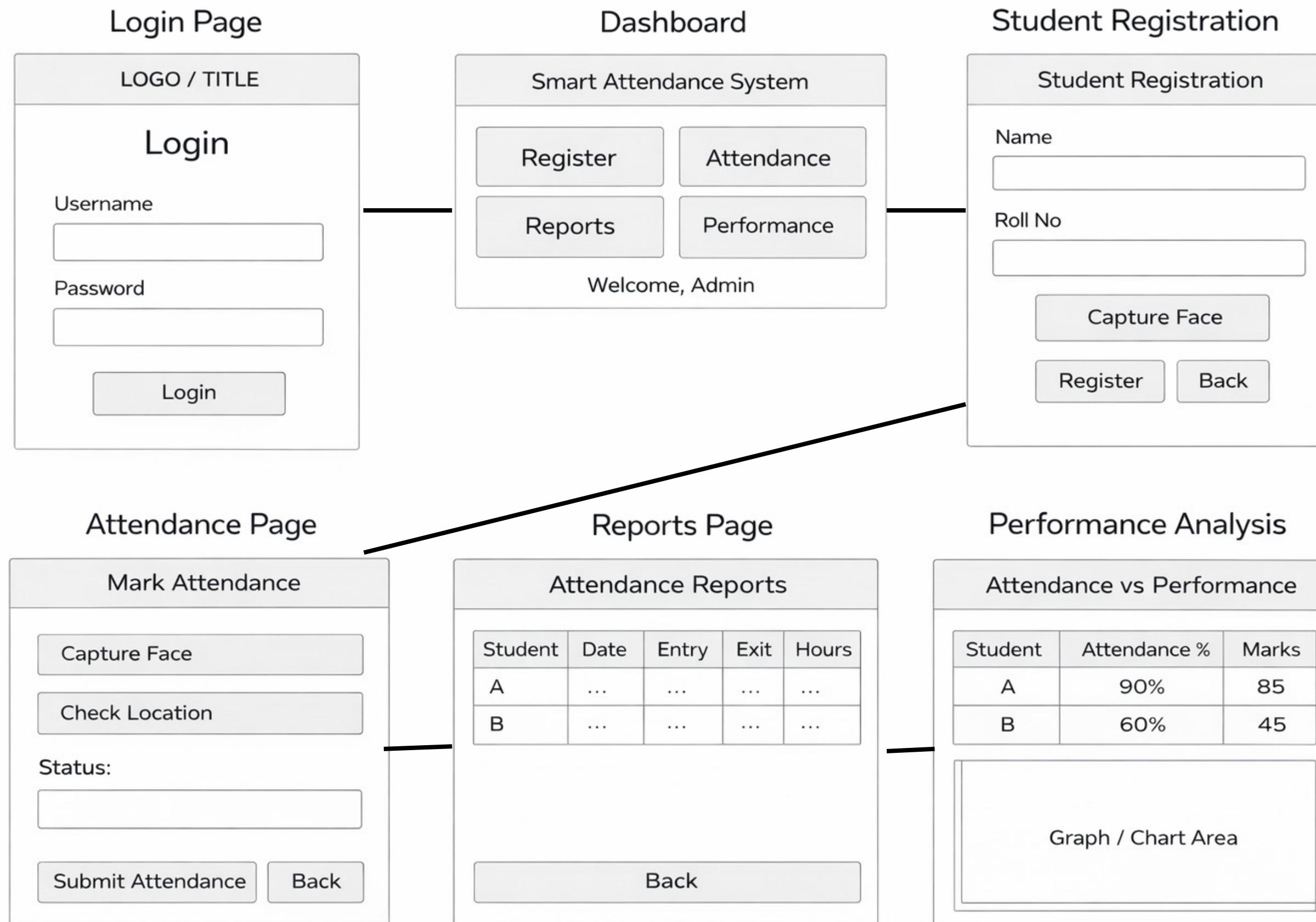
Use Case diagram



Process diagram



UI Wireframe



Thank
you

